

Empirical Industrial Organization

Chapter 6 Problem Set

Entry and Exit

What this packet contains

This PDF is a worked problem set built from the Chapter 6 lecture material on entry and exit. It is designed as a revision packet: each problem is followed immediately by a detailed solution.

The set covers:

- monopoly and oligopoly entry models,
- complete-information and incomplete-information games,
- heterogeneous fixed costs and market-level common shocks,
- ordered-response likelihoods, partial identification, and moment inequalities,
- airline-style entry applications, and
- policy interpretation of observed market structure.

Study note

Notation follows the lecture where practical. Some exercises use slightly simplified symbols so that the algebra is easier to read on the page.

Quick reference

Core notation

$v(x)$	Deterministic variable profit in the monopoly entry model.
F	Random fixed or sunk entry cost.
Ψ	Distribution function of fixed costs.
$v_y(x)$	Per-firm variable profit when y firms are active.
y_m	Observed number of firms in market m .
$v0_i(x_i)$	Firm i 's profit component when the rival does not enter.
$v1_i(x_i)$	Effect of rival entry on firm i 's profit; usually nonpositive.
$\mu(x), \delta(x)$	Entry probabilities tied to monopoly-like and duopoly-like thresholds.
K	Number of potential entrants in a market.
t_i	Entry threshold in the incomplete-information model.

Five model families

Setting	Observable	Main empirical idea
Monopoly entry	Observed entry rate	Need a normalization, known Ψ , known v , or shape restrictions such as homogeneity.
Homogeneous oligopoly	Number of firms	Use threshold structure $P\{Y \geq y x\} = \Psi(v_y(x))$; estimate as an ordered model.
Heterogeneous complete info	Entrant profile	Use outcome probabilities plus support and normalization arguments; sometimes only partial identification.
Multiple equilibria	Counts or profiles	Selection is unknown, so exploit moment equalities and inequalities instead of a full likelihood.
Incomplete information	Entrant profile	Solve threshold fixed points; then add a selection rule or impose a same-equilibrium assumption across markets.

Three questions to ask before estimating an entry model

- What is observed: a binary entry decision, a firm count, or the identity of entrants?
- Is the equilibrium uniquely predicted, or is partial identification more appropriate?
- What variation separates profits from costs: market size, potential entrants, firm asymmetries, or excluded shifters?

1 Foundations and market structure

Problem 1. Why entry and exit matter in empirical IO**Question**

Why is entry and exit a central topic in empirical industrial organization? Explain what these models identify and why the framework is broader than literal firm entry alone.

Detailed answer

Entry and exit models use observed market structure to recover a reduced-form profit function: variable profit minus the cost of entering. That object is valuable because it summarizes demand, cost, and conduct in a way that is directly informative about equilibrium market structure and producer surplus. The framework is broader than firm entry because the same logic applies to product launch, product withdrawal, store openings, route entry, and platform participation. In each case, the researcher studies a binary or discrete participation decision driven by net profits.

Problem 2. Competition, entry costs, and market size

Question

How do fixed or sunk entry costs interact with the mode of competition to shape equilibrium market structure?

Detailed answer

A market can support another entrant only when the entrant's variable profit is large enough to cover entry costs. If competition is tough, profits fall sharply as more firms enter, so the number of active firms grows slowly with market size. If entry costs are high, a market must be larger before another firm can profitably enter. This is the Sutton intuition: market structure is jointly determined by how hard firms compete after entry and by how costly it is to participate in the first place.

Problem 3. Exogenous versus endogenous entry costs

Question

Distinguish exogenous entry costs from endogenous entry costs. Why do endogenous entry costs tend to generate fewer but larger firms?

Detailed answer

Exogenous entry costs are costs that do not depend on the mode of entry; every potential entrant faces roughly the same fixed or sunk cost. Endogenous entry costs depend on how the entrant chooses to compete, for example through quality, advertising, or product design. As the number of incumbents rises, the entrant may need to spend more to achieve positive profit. That pushes equilibrium toward fewer active firms, while each surviving firm operates at larger scale because entry requires a bigger profitable niche.

Problem 4. Why profit functions can be identified without specifying conduct**Question**

The chapter emphasizes that entry models can identify profit functions without explicitly assuming a particular conduct model. Why is that useful?

Detailed answer

Observed entry decisions reveal whether net profit is above or below zero. Because net profit is already a reduced-form consequence of demand, cost, and conduct, the econometrician can learn about equilibrium profitability and market structure without separately modeling all of those primitives. This is useful when the data mainly record who enters rather than prices or quantities. It is also robust in the sense that a reduced-form profit function can still be informative even when the exact strategic details of post-entry competition are hard to pin down.

2 Monopoly entry

Problem 5. Threshold-crossing rule in the monopoly model

Question

In the monopoly entry model, profits are $\pi(x_i, F_i) = v(x_i) - F_i$. Derive the entry rule and the observed entry probability when F is independent of x with distribution Ψ .

Detailed answer

The entrant enters if and only if profit is nonnegative, so the decision rule is $F_i \leq v(x_i)$. If F is independent of x , the probability of entry at characteristics x is therefore

$$p(x) = P\{\text{entry} \mid x\} = P\{F \leq v(x)\} = \Psi(v(x)).$$

So the observed entry rate is a threshold-crossing probability: the latent variable-profit index $v(x)$ is compared with the random fixed cost F .

Problem 6. Why the monopoly model is not identified without restrictions

Question

Why are v and the distribution of F not separately identified in the basic monopoly entry model?

Detailed answer

Only the ordering of F relative to $v(x)$ matters for the entry decision. If $T(\cdot)$ is any strictly increasing transformation, then $F \leq v(x)$ is equivalent to $T(F) \leq T(v(x))$. That means the data cannot distinguish the original pair (v, Ψ) from the transformed pair $(T \circ v, \text{distribution of } T(F))$. Without a normalization or shape restriction, the model identifies at most the ranking of markets, not the absolute scale of profits and costs.

Problem 7. Identification when the fixed-cost distribution is known

Question

Suppose Ψ is known. Show how to identify the variable-profit function $v(x)$ from observed entry probabilities $p(x)$.

Detailed answer

From the threshold-crossing equation $p(x) = \Psi(v(x))$, apply the inverse cdf to both sides. This gives

$$v(x) = \Psi^{-1}(p(x)).$$

Economically, the observed entry rate tells you which quantile of the fixed-cost distribution a market reaches. Converting the probability into a quantile recovers the latent variable profit at that x .

Problem 8. Identification when variable profits are known

Question

Suppose $v(x)$ is known from outside information, such as independently identified demand and cost objects. Show how to identify Ψ .

Detailed answer

Take any value z in the range of v . Find an x such that $v(x) = z$. Then the observed entry probability at that x is

$$p(x) = P\{F \leq v(x)\} = P\{F \leq z\} = \Psi(z).$$

So one recovers the fixed-cost distribution by re-indexing observed entry probabilities using the known profit mapping $v(x)$. In compact notation, $\Psi(z) = p(v^{-1}(z))$.

Problem 9. Homogeneity and market-size identification

Question

Explain how the restriction $v(z, x_{\tilde{}}) = z v(x_{\tilde{}})$ together with a normalization $v(x_{\tilde{}}_0) = 1$ identifies both v and Ψ . Why can this restriction be economically plausible?

Detailed answer

At the normalized market type $x_{\tilde{}}_0$, profits satisfy $v(z, x_{\tilde{}}_0) = z$. Therefore

$$\Psi(z) = P\{F \leq z\} = P\{\text{entry} \mid z, x_{\tilde{}}_0\}.$$

This identifies the fixed-cost distribution directly from markets that differ only in size z . Once Ψ is known, any other market type $x_{\tilde{}}$ is identified by inverting $p(z, x_{\tilde{}}) = \Psi(z v(x_{\tilde{}}))$; equivalently, $v(x_{\tilde{}})$ is the scale factor that maps market size into the observed entry probability. The restriction is plausible when demand is proportional to population and marginal cost is constant, because then variable profit scales roughly linearly with market size.

3 Complete-information homogeneous oligopoly entry

Problem 10. Equilibrium condition with a common market-level fixed cost

Question

In the Bresnahan-Reiss style model, each market has a common fixed cost F_m and per-firm variable profit $v_y(x_m)$ when y firms operate. State the equilibrium condition for observing y firms.

Detailed answer

The equilibrium number of firms y is pinned down by the interval in which the common fixed cost falls:

$$v_y(x_m) \geq F_m > v_{y+1}(x_m).$$

The first inequality says the y -th firm can cover fixed cost, while the second says the next potential entrant cannot. Thus the sequence $v_1(x)$, $v_2(x)$, ... acts like a set of ordered profitability thresholds.

Problem 11. Probabilities of observing y firms or at least y firms

Question

Using the equilibrium condition above and a cdf Ψ for F , derive $P\{Y = y \mid x\}$ and $P\{Y \geq y \mid x\}$.

Detailed answer

The event $Y = y$ occurs exactly when F lies between $v_{y+1}(x)$ and $v_y(x)$. Hence

$$P\{Y = y \mid x\} = \Psi(v_y(x)) - \Psi(v_{y+1}(x)).$$

Similarly, observing at least y firms only requires that the y -th entrant be profitable, so

$$P\{Y \geq y \mid x\} = \Psi(v_y(x)).$$

These formulas are central because the cumulative probabilities map directly into profit thresholds.

Problem 12. Why the model is an ordered-response model

Question

Why can the homogeneous oligopoly entry model be estimated as an ordered-response model? What does ordered probit mean in this context?

Detailed answer

A single scalar unobservable, the market fixed cost F , is compared with an ordered list of cutoffs $v_1(x)$, $v_2(x)$, That is exactly the structure of an ordered discrete-choice model: different intervals of F correspond to different observed firm counts. If Ψ is assumed normal, the implied ordered-response model is an ordered probit. The econometrician then estimates the threshold structure through the likelihood of the observed number of firms.

Problem 13. Identification under a market-size shape restriction

Question

Suppose $v_y(z, x_{\tilde{}}) = z v_y(x_{\tilde{}})$ and $v_1(x_{\tilde{}}_0) = 1$. Outline the identification logic.

Detailed answer

First use markets of type $x_{\tilde{}}_0$ to recover the fixed-cost distribution:

$$\Psi(z) = P\{Y \geq 1 \mid z, x_{\tilde{}}_0\}.$$

Then use the cumulative probability for any y and any other market type $x_{\tilde{}}$:

$$P\{Y \geq y \mid z, x_{\tilde{}}\} = \Psi(z v_y(x_{\tilde{}})).$$

Because Ψ is now known, the researcher can invert the probability and divide by z to obtain $v_y(x_{\tilde{}})$. In words, market-size variation identifies the distribution, and cross-market composition identifies how profits differ across market types.

Problem 14. Numerical example with a uniform fixed-cost distribution**Question**

Suppose $F \sim \text{Uniform}[0,10]$, $v_1 = 8$, $v_2 = 5$, $v_3 = 2$, and $v_4 = 0$. Compute the probabilities of observing 0, 1, 2, and 3 firms. Also compute $P\{Y \geq 2\}$.

Detailed answer

Under a $\text{Uniform}[0,10]$ distribution, $\Psi(a) = a/10$ for a between 0 and 10. Therefore

$$P\{Y=0\} = 1 - \Psi(8) = 0.2$$

$$P\{Y=1\} = \Psi(8) - \Psi(5) = 0.8 - 0.5 = 0.3$$

$$P\{Y=2\} = \Psi(5) - \Psi(2) = 0.5 - 0.2 = 0.3$$

$$P\{Y=3\} = \Psi(2) - \Psi(0) = 0.2 - 0 = 0.2.$$

The cumulative probability of at least two firms is simply $\Psi(5) = 0.5$. This illustrates how the entry thresholds directly determine market-structure probabilities.

4 Complete-information heterogeneous oligopoly entry

Problem 15. Economic meaning of $v0_i$ and $v1_i$

Question

In the two-firm heterogeneous model, profits are $\pi_{im} = v0_i(x_{im}) + y_{jm} v1_i(x_{im}) - f_{im}$. What do $v0_i$ and $v1_i$ represent, and why are $v1_i$ and $v1_j$ typically nonpositive?

Detailed answer

The term $v0_i(x_{im})$ is firm i 's net variable-profit component when it is the only active firm. The term $v1_i(x_{im})$ is the effect on i 's profit of having rival j enter. Because a rival usually steals demand or intensifies competition, that incremental effect is typically zero or negative. Allowing $v1_i$ and $v1_j$ to differ admits asymmetric competition: firm j may hurt i more than i hurts j .

Problem 16. Pure-strategy equilibrium conditions for all four outcomes

Question

Write the pure-strategy Nash equilibrium conditions for outcomes (0,0), (1,0), (0,1), and (1,1) in the two-firm complete-information model.

Detailed answer

The conditions are:

$$(0,0): f_i \geq v0_i(x_i), f_j \geq v0_j(x_j)$$

$$(1,0): f_i \leq v0_i(x_i), f_j \geq v0_j(x_j) + v1_j(x_j)$$

$$(0,1): f_i \geq v0_i(x_i) + v1_i(x_i), f_j \leq v0_j(x_j)$$

$$(1,1): f_i \leq v0_i(x_i) + v1_i(x_i), f_j \leq v0_j(x_j) + v1_j(x_j).$$

Each condition states that active firms must find entry profitable given the rival's action, while inactive firms must find staying out optimal.

Problem 17. Tamer's identification logic under pure strategies

Question

Give the intuition behind Tamer's identification strategy for the heterogeneous two-firm complete-information model.

Detailed answer

Tamer starts from probabilities of observed equilibrium outcomes, especially the no-entry region. Under the normalization $v0_i(z_i, x_tilde_0) = z_i$ and $v0_j(z_j, x_tilde_0) = z_j$, the probability of no entry reveals the joint upper tail of the fixed-cost distribution:

$$P\{0,0 \mid z_i, x_tilde_0, z_j, x_tilde_0\} = P\{f_i \geq z_i, f_j \geq z_j\}.$$

Once the joint distribution is recovered, send z_j to $-\infty$ so firm j is effectively never profitable. That collapses the joint tail into a marginal tail for i , allowing $v0_i(x_i)$ to be identified. Similar arguments, using the competitive-effect terms, identify $v1_i$ and $v1_j$.

Problem 18. Berry's independent fixed-cost extension with two potential entrants

Question

Define $\mu(x) = P\{F_i < v_1(x)\}$ and $\delta(x) = P\{F_i < v_2(x)\}$. Suppose a market has exactly two potential entrants and $x_1 = x_2 = x$. Show how to identify $\mu(x)$ and $\delta(x)$ from $P\{0,0|x,x\}$ and $P\{1,1|x,x\}$.

Detailed answer

With independent fixed costs, the no-entry probability factorizes:

$$P\{0,0|x,x\} = [1 - \mu(x)]^2.$$

Therefore

$$\mu(x) = 1 - \sqrt{P\{0,0|x,x\}}.$$

Likewise, observing both firms enter implies

$$P\{1,1|x,x\} = \delta(x)^2, \text{ so } \delta(x) = \sqrt{P\{1,1|x,x\}}.$$

These two objects summarize the profitability cutoffs for monopoly-like entry and duopoly entry when there are exactly two potential entrants.

Problem 19. Why the number of potential entrants helps identification

Question

Why is observing the number of potential entrants useful in heterogeneous entry models?

Detailed answer

Variation in the number of potential entrants generates different equilibrium probabilities even when market characteristics are otherwise similar. That creates extra equations linking observed outcomes to the same underlying profit and cost primitives. In airline applications, for example, knowing how many carriers could plausibly serve a route helps separate firm-level heterogeneity from common market conditions. Without this variation, many combinations of costs and competitive effects can fit the same realized entrant set.

Problem 20. Market-level correlated fixed costs and the role of K

Question

Suppose there is a common market shock ε_m that takes values 0 and 1 with probabilities λ and $1-\lambda$. Let $a = 1 - \mu(x,0)$ and $b = 1 - \mu(x,1)$. Explain why observing $P\{\text{no entry} \mid K\}$ for different numbers of potential entrants K helps identify λ , a , and b .

Detailed answer

When all K potential entrants have the same observable type x , the no-entry probability is

$$P\{0, \dots, 0 \mid K\} = \lambda a^K + (1-\lambda) b^K.$$

This is a finite-mixture expression. With $K = 1, 2, 3$ one obtains three equations in the three unknowns λ , a , and b . Intuitively, K changes how strongly the common shock propagates into the probability that nobody enters. That extra curvature in K allows the econometrician to distinguish true heterogeneity across market states from ordinary independent fixed-cost dispersion.

Problem 21. Numerical recovery in the binary common-shock model

Question

Suppose the observed no-entry probabilities for otherwise identical markets are $P_1 = 0.55$ when $K=1$, $P_2 = 0.325$ when $K=2$, and $P_3 = 0.2035$ when $K=3$. Recover λ , $a = 1 - \mu(x,0)$, and $b = 1 - \mu(x,1)$.

Detailed answer

The model implies $P_K = \lambda a^K + (1-\lambda) b^K$. From $K=1$ and $K=2$, note that $a = 0.7$ and $b = 0.4$ satisfy

$$0.55 = \lambda(0.7) + (1-\lambda)(0.4) \text{ only if } \lambda = 0.5,$$

and also

$$0.325 = 0.5(0.7^2) + 0.5(0.4^2) = 0.245 + 0.08.$$

Checking $K=3$ gives

$$0.2035 = 0.5(0.7^3) + 0.5(0.4^3) = 0.1715 + 0.032.$$

So the recovered parameters are $\lambda = 0.5$, $a = 0.7$, and $b = 0.4$. Equivalently, $\mu(x,0) = 0.3$ and $\mu(x,1) = 0.6$.

5 Airline applications and unique predictions

Problem 22. Interpreting Berry's airline profit function

Question

In Berry's airline application, the market is a city-pair route and firm i 's profit takes the form $\pi_{im} = v_m(N_m) - f_{im}$. Interpret the components

$$v_m(N) = x_m'\beta + h(\delta, N) + \rho u_m, \quad f_{im} = z_{im}'\alpha + \sigma u_{im}.$$

Detailed answer

The vector x_m contains market-level observables such as route characteristics that shift common profitability. The function $h(\delta, N)$, often written as $-\delta \ln N$, captures how profitability declines as more firms serve the route, so δ measures the intensity of crowding or competition. The term z_{im} contains firm-specific observables that shift costs or route suitability. Finally, ρu_m is a market-level unobservable common to all firms in a route, while σu_{im} is a firm-specific unobservable, with the scale normalization chosen so their sum has unit variance.

Problem 23. Why a unique number of entrants is valuable

Question

Why does Berry emphasize conditions under which the equilibrium number of entrants is unique?

Detailed answer

If the number of entrants is uniquely predicted by observables and shocks, the econometrician can write a standard likelihood for the observed counts and estimate the model directly. Multiplicity breaks that likelihood because the same fundamentals can generate several admissible outcomes. Berry therefore imposes assumptions that restore uniqueness, such as no firm-level unobserved heterogeneity ($\rho=1$), no market-level unobserved heterogeneity ($\rho=0$), or a predetermined order of entry. The benefit is tractability; the cost is that the model becomes less general.

Problem 24. Unique number of entrants versus unique identity of entrants**Question**

Why is a unique number of entrants weaker than a unique identity of entrants?

Detailed answer

It is possible for the model to pin down that exactly one or exactly two firms enter while still leaving open which firms those are. This happens when several entrant profiles produce the same count. For likelihood-based estimation of counts, uniqueness of the number of entrants is enough. But if the data record the identity of entrants, then remaining multiplicity in the profile matters and must be handled explicitly.

6 Multiple equilibria and moment inequalities

Problem 25. Multiple equilibria versus multiple predictions

Question

Distinguish multiple equilibria from multiple predictions in entry models.

Detailed answer

Multiple equilibria means that the same fundamentals support more than one Nash equilibrium. Multiple predictions means that the same parameter vector does not map into a unique prediction for the observable outcome. The two ideas are related but not identical: a model can have multiple equilibria over entrant identities while still making a unique prediction for the number of entrants. That distinction is crucial for deciding whether one can write a likelihood for counts, for identities, or only bounds.

Problem 26. Why the count of entrants may be identified when the entrant profile is not

Question

In the bivariate complete-information game with multiplicity, why might the likelihood of the number of entrants be available even when the likelihood of the entrant profile is not?

Detailed answer

In the multiplicity region, the outcomes $(1,0)$ and $(0,1)$ can both be equilibria, so the model cannot tell which firm enters. But both outcomes imply the same count: one entrant. Therefore the event $Y = 1$ may still be uniquely predicted even when the exact strategy profile is not. This is why some models can be estimated with data on firm counts but not with richer data on entrant identities unless an equilibrium-selection rule is imposed.

Problem 27. Why $P\{0,0|x\}$ and $P\{1,1|x\}$ are point-identified

Question

In Tamer's pure-strategy analysis, why are the probabilities of observing (0,0) and (1,1) written as equalities, while the off-diagonal outcomes are bounded?

Detailed answer

The regions of the unobservable space that generate (0,0) and (1,1) are unique-equilibrium regions: whenever those inequalities hold, the corresponding outcome must occur. In contrast, the regions that support (1,0) or (0,1) include both unique-equilibrium regions and an overlap region with multiple equilibria. Because the econometrician does not know how the economy selects within that overlap, only lower and upper bounds can be written for the off-diagonal probabilities.

Problem 28. Reading moment-inequality bounds

Question

Suppose a candidate parameter vector implies $H_{\text{lower}}(0,1|x) = 0.15$ and $H_{\text{upper}}(0,1|x) = 0.27$. Is the parameter vector admissible if the data show $P\{0,1|x\} = 0.22$? What if the data show 0.31?

Detailed answer

The parameter vector is admissible when the observed probability lies inside the model-implied interval. Thus 0.22 is admissible because $0.15 \leq 0.22 \leq 0.27$. But 0.31 is not admissible because it exceeds the upper bound. In the GMM formulation, the first case makes the inequality moments equal to zero, while the second case produces a positive violation through the $\max\{P - H_{\text{upper}}, 0\}$ term.

7 Incomplete-information heterogeneous entry

Problem 29. Why equilibrium strategies are thresholds

Question

In the incomplete-information two-firm model, why is it natural for equilibrium strategies to take threshold form $y_i = 1\{f_i \leq t_i\}$?

Detailed answer

For a given belief about the rival's behavior, firm i 's expected payoff from entering falls one-for-one with its own fixed cost f_i . That monotonicity implies a cutoff rule: if entry is profitable at some cost level, it is also profitable at any lower cost. Hence the optimal strategy can be summarized by a threshold t_i such that all types with f_i below the threshold enter and all higher-cost types stay out.

Problem 30. Deriving the threshold fixed-point equations

Question

Starting from the expected payoff expression, derive the threshold equation for firm i .

Detailed answer

If firm j uses threshold t_j , let $G_i^j(t_j)$ be firm i 's belief that j enters. At the cutoff type $f_i = t_i$, firm i must be indifferent between entering and staying out:

$$(v_{0i} - t_i)(1 - G_i^j(t_j)) + (v_{0i} + v_{1i} - t_i) G_i^j(t_j) = 0.$$

Collecting terms yields the compact fixed-point condition

$$t_i = v_{0i}(x_i) + v_{1i}(x_i) G_i^j(t_j).$$

An analogous equation holds for j . Because each threshold depends on the other firm's threshold through beliefs, the system can have more than one solution.

Problem 31. Role of common priors and beliefs

Question

Interpret $G_i^j(t_j)$. How does the meaning change when fixed costs are independent versus correlated?

Detailed answer

The object $G_i^j(t_j)$ is firm i 's belief that rival j 's fixed cost is low enough to induce entry, given i 's information. If fixed costs are independent and firms share a common prior, then $G_i^j(t_j)$ is simply j 's unconditional entry probability $F_j(t_j)$. If fixed costs are correlated, i 's own type contains information about j 's type, so the belief becomes a conditional probability. In that case, the slope of i 's best response reflects both strategic interaction and informational correlation.

Problem 32. Numerical symmetric-threshold example

Question

Assume f_i and f_j are independent Uniform[0,1], $v0_i = v0_j = 0.8$, and $v1_i = v1_j = -0.4$. Find the symmetric equilibrium threshold t .

Detailed answer

With independent Uniform[0,1] costs, the probability that the rival enters equals its threshold, so in a symmetric equilibrium $G(t) = t$. The threshold equation is therefore

$$t = 0.8 - 0.4 t.$$

Solving gives $1.4 t = 0.8$, so

$$t = 4/7 \approx 0.571.$$

Thus each firm enters when its fixed cost is at most about 0.571, and the equilibrium entry probability of each firm is also about 57.1 percent.

8 Synthesis and review

Problem 33. What patterns in the data distinguish high entry costs, tough competition, and common shocks?

Question

Suppose you observe that the number of firms rises only slowly with market size and incumbent firms remain large. Give two economic interpretations of this pattern and describe what additional evidence would help you separate them.

Detailed answer

One interpretation is tough post-entry competition: as more firms enter, variable profit per firm falls quickly, so only a few firms can survive even in large markets. A second interpretation is endogenous sunk entry costs: entrants must spend heavily on quality or positioning to compete, so equilibrium again features a small number of large firms. To separate them, the researcher would look for evidence on quality investment, product positioning, or entrant-specific costs; variation in the number of potential entrants; and whether entry thresholds shift mostly with market size or with strategic crowding. Common market-level shocks are another possibility, so route- or market-level covariates and repeated observations are also important for ruling out latent demand conditions.

Selected references from the chapter

Bresnahan & Reiss (1991); Berry (1992); Berry & Tamer (2006); Tamer (2003); Ciliberto & Tamer (2009); Sutton (2007).